Lecture 9 Second Order Nonhomogeneous ODEs

Lecture 9 Second O...

Engineering Mathematics III

ENGE702

Lecture 9: Second Order Nonhomogeneous ODEs

Second order ODEs

In this lecture we will look at methods of solving second order nonhomogeneous linear ODEs. That is, ODEs of the form:

$$y'' + p(x)y' + q(x)y = r(x)$$

where $r(x) \neq 0$

It turns out that finding a solution to this type of differential equation is very much related to solving the corresponding homogeneous equation (ie with $r(x) = 0$).

The solution will in fact be the solution to the corresponding homogeneous equation, plus a particular solution to the original, nonhomogeneous equation.

Second order ODEs

Given a second order linear differential equation

$$y'' + p(x)y' + q(x)y = r(x),$$

the general solution will be of the form

$$y = y_p + y_h$$

where:

y_h is the general solution to the homogeneous equation $y'' + p(x)y' + q(x)y = 0$, and

y_p is any particular solution to the original equation $y'' + p(x)y' + q(x)y = r(x)$.

For the homogenous 2nd linear ODE

$$y'' + ay' + by = 0$$

we find the characteristic equation

$$\lambda^2 + a\lambda + b = 0$$

and solve for the roots λ_1, λ_2

and get the general solution

$$y(x) = c_1 \cdot e^{\lambda_1 x} + c_2 \cdot e^{\lambda_2 x}$$

Extend to the inhomogenous 2nd order linear ODE

$$y'' + ay' + by = \boxed{r(x)}$$

The general solution of the inhomogenous 2nd order linear ODE has the form:

$$y = y_p + y_h.$$

y_h is the general solution for the homogenous problem which we know how to solve

y_p is any particular soluton to the inhomogenous problem.

We use a so-called method of undetermined coefficients to find a particular solution.

Second order ODEs

Now we have two distinct jobs to do to solve the nonhomogeneous equation.

In the previous lecture, we saw how to solve second order homogeneous linear differential equations with constant coefficients. We will today cover a method to find a particular solution to a second order nonhomogeneous linear differential equation with constant coefficients.

Therefore, we will concentrate from here on **second order nonhomogeneous linear differential equations with constant coefficients**

We already know how to solve second order homogeneous differential equations with constant coefficients, we will see now how to find a particular solution using a method known as the **method of undetermined coefficients**.

Second order ODEs

The method of undetermined coefficients works by 'guessing' a solution (with some unknown coefficients) based on $r(x)$, and then putting this back into the equation to work out what the coefficients should be.

The table of 'guesses' for common forms of $r(x)$ is as follows:

Term in $r(x)$	Choice for y_p
$ke^{\gamma x}$	$Ke^{\gamma x}$
kx^n	$K_n x^n + K_{n-1}x^{n-1} + \cdots + K_1 x + K_0$
$k\cos(\omega x)$	$K_1\cos(\omega x) + K_2\sin(\omega x)$
$k\sin(\omega x)$	$K_1\cos(\omega x) + K_2\sin(\omega x)$
$ke^{\alpha x}\cos(\omega x)$	$e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$
$ke^{\alpha x}\sin(\omega x)$	$e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$

Second order ODEs

Using the table, we can choose an appropriate y_p using the following rules:

- **Basic Rule:** If $r(x)$ is one of the functions in the first column of the table on the previous slide, choose the corresponding y_p, then determine the coefficients by inserting y_p and its derivatives into the differential equation, and equating the sides.
- **Sum Rule:** If $r(x)$ is a sum of functions in the first column of the table on the previous slide, choose y_p to be the sum of the corresponding functions in the table, and then determine the coefficients by inserting y_p and its derivatives into the differential equation, and equating the sides.
- **Modification Rule:** If a term in the choice for y_p is a solution of the corresponding homogeneous ODE, then multiply this term by x (or x^2 if the solution is a double root).

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + y' - 2y = \sin(x).$$

$r(x) \neq 0$

Example: Solve $y''+y'-2y=\boxed{\sin(x)}$

$y = y_h + y_p$

For y_h, we solve $y''+y'-2y=0$

$\Rightarrow \lambda^2+\lambda-2=0$

$\lambda = \dfrac{-1\pm\sqrt{1^2-4\times(-2)}}{2} = \dfrac{-1\pm 3}{2}$

$\lambda_1 = 1$ and $\lambda_2 = -2$.

So $y_h = c_1 e^{x} + c_2 e^{-2x}$.

For y_p, we choose $y_p = K_1\cos(x) + K_2\sin(x)$

substitute into ODE

$y''+y'-2y=\sin(x)$

$y_p' = -K_1\sin(x) + K_2\cos(x)$

$y_p'' = -K_1\cos(x) - K_2\sin(x)$

$-K_1\cos(x) - K_2\sin(x) - K_1\sin(x) + K_2\cos(x) - 2K_1\cos(x) - 2K_2\sin(x) = \sin(x)$

$\underbrace{(-3K_1+K_2)}_{0}\cos(x) + \underbrace{(-3K_2-K_1)}_{1}\sin(x) = 1\sin(x) + 0\cos(x)$

$-3K_1+K_2=0 \Rightarrow K_2 = 3K_1$

$-3K_2-K_1=1 \Rightarrow -10K_1 = 1 \Rightarrow K_1 = -\frac{1}{10}$

$K_2 = -\frac{3}{10}$

So $y_p = -\frac{1}{10}\cos(x) - \frac{3}{10}\sin(x)$

Hence, the general solution is $y = y_p + y_h$

$= -\frac{1}{10}\cos(x) - \frac{3}{10}\sin(x) + c_1e^{x} + c_2e^{-2x}$.

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + y' - 2y = \sin(x).$$

The solution will be the sum of a sum of y_h and y_p.

From a previous lecture we know the solution of the corresponding homogeneous equation is $y_h = c_1 e^x + c_2 e^{-2x}$.

Now we need to find a particular solution. From the table, we see that the suggested choice for y_p is $y_p = K_1 \cos(x) + K_2 \sin(x)$. Then $y_p' = K_2 \cos(x) - K_1 \sin(x)$, and $y_p'' = -K_1 \cos(x) - K_2 \sin(x)$. So

$$\begin{aligned}
&y_p'' + y_p' - 2y_p \\
&= -K_1 \cos(x) - K_2 \sin(x) + K_2 \cos(x) - K_1 \sin(x) - 2K_1 \cos(x) - 2K_2 \sin(x) \\
&= (K_2 - 3K_1) \cos(x) + (-K_1 - 3K_2) \sin(x)
\end{aligned}$$

Second order ODEs

Since this must equate to $0\cos(x) + 1\sin(x)$ (which is $r(x)$ - the right hand side in our equation), we conclude that

$$K_2 - 3K_1 = 0$$

and

$$-K_1 - 3K_2 = 1$$

Solving this we get

$$K_1 = -\frac{1}{10} \quad \text{and} \quad K_2 = -\frac{3}{10}$$

Then our particular solution is $y_p = -\frac{1}{10}\cos(x) - \frac{3}{10}\sin(x)$, so our general solution is then

$$y = y_h + y_p = c_1 e^x + c_2 e^{-2x} - \frac{1}{10}\cos(x) - \frac{3}{10}\sin(x)$$

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' - y' - 2y = 2e^{-x}.$$

Second order ODEs

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$$y'' - y' - 2y = 2e^{-x}.$$

Using what we know about solving homogeneous equations, we can see that $y_h = c_1 e^{-x} + c_2 e^{2x}$.

Now we need to find a particular solution. From the table, we see that the suggested choice for y_p is $y_p = Ce^{-x}$. This is the same as one of our solutions, so instead we use $y_p = Cxe^{-x}$.

Then $y_p' = C(1-x)e^{-x}$ and $y_p'' = C(x-2)e^{-x}$. So

$$\begin{aligned}
&y_p'' - y_p' - 2y_p \\
&= C(x-2)e^{-x} - C(1-x)e^{-x} - 2Cxe^{-x} \\
&= -3Ce^{-x}
\end{aligned}$$

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Equating this to $r(x)$ we get $-3C = 2$, so $C = -\frac{2}{3}$.

Then our general solution is

$$y = c_1 e^{-x} + c_2 e^{2x} - \frac{2}{3} x e^{-x}$$

Exercises

Find the solution to the following differential equations:

- $y'' + y = e^{2x}$
- $y'' - y = x^2$

Text Book References

Chapter 2.7